The empirical recurrence for OEIS A235021: recovering a conjecture that is not written down

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Abstract

OEIS A235021 records "Empirical recurrence of order 63", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 750$ vertices and satisfies some monic recurrence of order at most S; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 63, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 63 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A235021 is "Number of (n+1) X (3+1) 0..5 arrays with every 2 X 2 subblock having its diagonal sum differing from its antidiagonal sum by 3, with no adjacent elements equal (constant-stress tilted 1 X 1 tilings).". It has offset 1 and begins

$$1248, 4312, 14800, 51220, 177520, 617600, 2152744, 7527124, \ \ldots$$

The entry states:

> Empirical recurrence of order 63 (see link above).

As of the "Last modified" line on the live entry (Jun 20 2022 00:25:12, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 750$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S =$ 750.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1500$ exact terms of the model returns order 63 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{63} c_i \, a(n-i),$$

c_1	8	c_{17}	1137439498832	c_{33}	52350881921137615	c_{49}	29331743328691200
c_2	106	c_{18}	−1041251531813	c_{34}	−45740935581224310	c_{50}	−8938846384699392
c_3	−977	c_{19}	−8171747558034	c_{35}	−92509294783749752	c_{51}	−10668562901262336
c_4	−4970	c_{20}	8271283993687	c_{36}	73914989930416554	c_{52}	2678600072261632
c_5	55453	c_{21}	48462092734085	c_{37}	137635819564403616	c_{53}	2970941717741568
c_6	132538	c_{22}	−51673300705763	c_{38}	−99311923771234712	c_{54}	−600100131487744
c_7	−1946194	c_{23}	−239215461461196	c_{39}	−171784238193993832	c_{55}	−610219397545984
c_8	−2069793	c_{24}	259861256550176	c_{40}	110612738550641472	c_{56}	95748185784320
c_9	47420231	c_{25}	988919110810534	c_{41}	178962017190568560	c_{57}	87447199285248
c_{10}	13693293	c_{26}	−1066928354991435	c_{42}	−101655262691469488	c_{58}	−10051209658368
c_{11}	−854081181	c_{27}	−3438970734353124	c_{43}	−154593306697116000	c_{59}	−8003786899456
c_{12}	176231878	c_{28}	3608880597413366	c_{44}	76583456381335904	c_{60}	598762061824
c_{13}	11825375044	c_{29}	10088485972487318	c_{45}	109787174738198400	c_{61}	401109680128
c_{14}	−6361371466	c_{30}	−10114816465209382	c_{46}	−46876141199582784	c_{62}	−14092861440
c_{15}	−129272614664	c_{31}	−25001951822619030	c_{47}	−63391369550366720	c_{63}	−8053063680
c_{16}	98726269276	c_{32}	23570886563592099	c_{48}	23033610297180416		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 63, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $63 + 4$ terms removed returns order 63 as well, so $\deg q_{\min} = 63 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 21 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $63 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 63 that this sequence satisfies — and that family turns out to have exactly one member.

References

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